

Algebra 1 Unit 5 Assessment Guide

General Information	Writing equations for a real-world context is difficult for students. Students who understand how to write one equation can often transfer that understanding to writing a system of equations. In grades 6-8, students have built a foundation of writing equations from a real-world context. Students who struggle with writing equations may benefit from reviewing these lessons in grades 6 and 7: 6th Grade, Unit 12, Lesson 4 and 7th Grade, Unit 10, Lessons 8-10. Sometimes students struggle with a context because they do not have the life experience. When remediating, help students “experience” the context as much as possible. Using manipulatives to explain the “story” of the context may help students learn to think about what in the context varies and needs to have a variable and what in the context stays constant. Solving equations and rewriting a formula for a variable are related. Students often struggle with understanding how to get a variable by itself. In units 3 and 4, students were rearranging equations of lines. These skills should be helpful for students who struggle with the benchmark MA.912.AR.1.2. This skill can also be used when solving a system of equations where students have to rearrange the system to solve using elimination. Students who need support on MA.912.AR.1.2 may benefit from practicing solving systems where various rearrangements are necessary. Solving equations and solving a system of equations are also related. Students who can solve an equation with variables on both sides are essentially solving a system by substitution. The questions for MA.912.AR.9.1 that require the student solve algebraically do not state a method. How a student solves a system of equations is not what is assessed on a large-scale assessment, so the method is left to the student to decide.
Calculator Information	The questions on the unit assessment are calculator.
Total Recommended Points	The total recommended points in this guide are 100 points.

Question	1a	Benchmark(s)	MA.912.AR.2.1
$300 - 4h = 206.4$		Recommended Scoring (4 Total Points)	Students may inadvertently use the wrong variable. Consider a 1-point deduction for this mistake. Consider deducting 3 points if students write $300 + 4h = 206.4$.
Question	1b	Benchmark(s)	MA.912.AR.2.1
23.4		Recommended Scoring (8 Total Points)	Consider reviewing work for minor errors that students carry through the problem. Students who make such errors may be showing their knowledge of the benchmark, so partial credit should be considered. It is suggested that minor errors have no more than a 2-point deduction. Consider deducting 4 points for each algebraic error. If students correctly solve the equation they wrote in 1a, they should not lose points.

Question	2	Benchmark(s)	MA.912.AR.2.1
	$9.64h + 82.35 = 14.86h + 17.10$	Recommended Scoring (4 Total Points)	Students may inadvertently use the wrong variable. Consider a 1-point deduction for this mistake. Consider deducting 2 points for students who have $9.64h + 82.35$ and $14.86h + 17.10$ but do not set the equations equal to each other.
Question	3a	Benchmark(s)	MA.912.AR.2.1
	$598.74 = 183 + 0.39(m - 120)$	Recommended Scoring (4 Total Points)	Students may also correctly write $598.74 = 0.39(m - 120) + 183$ or equivalent. Students may inadvertently use the wrong variable. Consider a 1-point deduction for this mistake. Consider deducting 3 points if students write $598.74 = 183 + 0.39n - 120$ or equivalent.

Question	3b	Benchmark(s)	MA.912.AR.2.1
	1,186	Recommended Scoring (8 Total Points)	Consider reviewing work for minor errors that students carry through the problem. Students who make such errors may be showing their knowledge of the benchmark so partial credit should be considered. It is suggested that minor errors have no more than a 2-point deduction. Consider deducting 4 points for each algebraic error. If students correctly solve the equation they wrote in 3a, they should not lose points.
Question	4	Benchmark(s)	MA.912.AR.1.2
	$H = \frac{T - Gx}{y}$	Recommended Scoring (6 Total Points)	Consider deducting 2 points for each algebraic error. Equivalent answers with H isolated should be accepted.
Question	5	Benchmark(s)	MA.912.AR.1.2
	$d = \frac{300}{11} \cdot \frac{F}{F - 13}$	Recommended Scoring (6 Total Points)	Consider deducting 2 points for each algebraic error. Equivalent answers with d isolated should be accepted.

Question	6	Benchmark(s)	MA.912.AR.9.1
a. The system $\begin{cases} y - 3 = 12(4 - x) \\ y = 3(-4x + 3) \end{cases}$ b. The system $\begin{cases} y = 10 - 2x \\ 6x + 3y = 30 \end{cases}$ c. The system $\begin{cases} 4x + y = -16 \\ 2x = -5y - 26 \end{cases}$	☐ has no solution. ☐ has one solution. ☐ has an infinite number of solutions. ☐ has no solution. ☐ has one solution. ☐ has an infinite number of solutions. ☐ has no solution. ☐ has one solution. ☐ has an infinite number of solutions.	Recommended Scoring (15 Total Points)	Suggested points is 5 points for each statement.

Question	7	Benchmark(s)	MA.912.AR.9.1
	9	Recommended Scoring (8 Total Points)	Students may show their work in a variety of ways. Consider reviewing work for minor errors that students carry through the problem. Students who make such errors may be showing their knowledge of the benchmark, so partial credit should be considered. It is suggested that minor errors have no more than a 2-point deduction. Consider deducting 4 points for each algebraic error.
Question	8	Benchmark(s)	MA.912.AR.9.1
	No Solution	Recommended Scoring (8 Total Points)	Students may show their work in a variety of ways. Consider reviewing work for minor errors that students carry through the problem. Students who make such errors may be showing their knowledge of the benchmark, so partial credit should be considered. It is suggested that minor errors have no more than a 2-point deduction. Consider deducting 4 points for each algebraic error.

Question	9	Benchmark(s)	MA.912.AR.9.1
	(3, 2)	Recommended Scoring (8 Total Points)	Students may show their work in a variety of ways. Consider reviewing work for minor errors that students carry through the problem. Students who make such errors may be showing their knowledge of the benchmark, so partial credit should be considered. It is suggested that minor errors have no more than a 2-point deduction. Consider deducting 4 points for each algebraic error.
Question	10	Benchmark(s)	MA.912.AR.9.1
	(7, -4)	Recommended Scoring (8 Total Points)	Students may show their work in a variety of ways. Consider reviewing work for minor errors that students carry through the problem. Students who make such errors may be showing their knowledge of the benchmark, so partial credit should be considered. It is suggested that minor errors have no more than a 2-point deduction. Consider deducting 4 points for each algebraic error.

Question	11	Benchmark(s)	MA.912.AR.9.1
	$(-6, 1)$	Recommended Scoring (8 Total Points)	Students may show their work in a variety of ways. Consider reviewing work for minor errors that students carry through the problem. Students who make such errors may be showing their knowledge of the benchmark, so partial credit should be considered. It is suggested that minor errors have no more than a 2-point deduction. Consider deducting 4 points for each algebraic error.
Question	12	Benchmark(s)	MA.912.AR.9.1
	$(-2, -6)$	Recommended Scoring (5 Total Points)	No partial work is suggested.